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GR Jai Pradeep
Under Graduate,
Department of Agriculture, SRM College of Agricultural Sciences, SRM Institute of Science and Technology, Chengalpattu, Tamil Nadu, India

Dr. Sivakumar K
Assistant Professor,
Department of Agronomy, SRM College of Agricultural Sciences, SRM Institute of Science and Technology, Chengalpattu, Tamil Nadu, India.

BT Suresh Kumar
Post Graduate Research Scholar,
Department of Agronomy, SRM College of Agricultural Sciences, SRM Institute of Science and Technology, Chengalpattu, Tamil Nadu, India

Corresponding Author:
GR Jai Pradeep
Under Graduate,
Department of Agriculture, SRM College of Agricultural Sciences, SRM Institute of Science and Technology, Chengalpattu, Tamil Nadu, India

The role of IoT in sustainable and profitable farming

GR Jai Pradeep, Sivakumar K and BT Suresh Kumar

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Abstract
In this context, the use of advanced technologies like smart sensors combined with the Internet of Things (IoT) could greatly boost agricultural output and reduce economic losses. Research around the world has shown that integrating IoT and smart sensors effectively monitors environmental factors critical to crop growth, such as moisture, humidity, temperature, and soil composition. These systems can also track greenhouse gases like carbon dioxide and methane etc. Smart farming technology can even measure nitrogen levels in soil, helping farmers determine the right amount of fertilizer for their crops. IoT devices and unmanned aerial vehicles (drones) are also useful for accurately detecting pest infestations and related diseases in crops. While smart farming has great potential for the future, it does face challenges, including high setup costs, data security concerns, and a lack of digital knowledge among farmers. To make IoT-driven farming more accessible, economic support, data protection measures, and improved digital education for farmers would be beneficial.

Keywords: Internet of Things (IoT), irrigation, soil sensors, drones

Introduction

What is IoT?

The Internet of Things (IoT) refers to connecting physical objects that have built-in electronics, allowing them to communicate and interact with each other or their surroundings. In the future, IoT technology will provide advanced services that will greatly impact how people live their daily lives. The Internet of Things (IoT) is made up of smart devices that talk to each other and work together with other machines, objects, surroundings, and systems. (Karimi *et al.*, 2013) [8]. It is already making a difference in areas like medicine, energy, gene treatments, farming, smart cities, and smart house hold works. IoT is one of the key technologies behind "Industry 4.0." Its main purpose is to connect people with machines and smart technology (Gazis, 2021) [6].

The Internet of Things is a system of physical devices that can share data with each other without needing humans to be involved. The term was first used by computer scientist Kevin Ashton in 1999. IoT devices are not just computers or machines; they can be any physical object with a sensor and a unique ID. The main aim of IoT is to create devices that can report information on their own and communicate with each other and users in real time.

IOT in Agriculture

The Internet of Things (IoT) in agriculture shows how connected devices and sensors are used to collect, track, and send data in real time to improve farming practices. Farmers use IoT in new and smart ways to help grow more crops and improve productivity. (Farooq *et al.*, 2021) [5]. From sensors that measure soil moisture to drones, these devices help automate tasks like watering crops, checking crop health, and managing livestock. Today, using IoT sensors, mobile devices, cloud services, and data analysis together is very important for modern, precise farming (Senapaty, M.K *et al.*, 2023) [16]. By using accurate data, farmers can lower production costs, reduce waste, and grow better crops.

- Irrigation management.
- Soil sensors.
- Drones.
- Livestock management.
- pest management.

Irrigation Management

Traditional water management involves checking things manually, making rough guesses, and waiting for problems to happen. However, smart systems using IoT (Internet of Things) technology completely change this approach.

An IoT-based water management system uses smart sensors, meters, and automated controls to constantly monitor water usage, find problems, and improve how water is distributed. These systems help local governments and businesses make better decisions quickly by not just collecting data, but also analyzing it and taking action based on it. A smart irrigation system uses collected data to predict how much water crops need and then automatically controls the water flow without needing any human help (Nigussie *et al.*, 2020) [10].

Soil Sensors

Soil temperature is important for plant growth because it affects root development, breathing, and the breakdown of nitrogen in the soil. Carefully tracking soil moisture is important for supporting farming that's good for the environment (Wang *et al.*, 2023) [18]. IoT sensors can estimate soil temperature by measuring air temperature and other factors, but the most accurate way is to use a probe placed directly in the soil.

To get the best readings, multiple probes can be placed at different depths, depending on the plant's root system. For measuring surface soil temperature, a different type of IoT sensor that uses infrared (IR) technology can be used. Keeping track of soil nutrients is very useful, but it can't be done well without accurate soil maps (Othaman, N.N.C *et al.*, 2021) [11]. With the help of sensors, connectivity tools, and data analysis, farmers can now keep track of soil health in real-time. We can find out what minerals are in the soil by testing it, either in a lab or with sensors (Senapaty *et al.*, 2023) [16]. This leads to better crop production, more efficient use of resources, and more sustainable farming. IoT-based soil monitoring systems are important for checking key soil factors like moisture, temperature, pH, and nutrients, which are all crucial for managing farming effectively.

Drones

Traditional farming methods usually require a lot of manual work, using a one-size-fits-all approach, and solving problems as they come up. These methods take a lot of time, use many resources, and can be inefficient. On the other hand, drones used in farming allow for more efficient, data-driven, and targeted farming practices. They give farmers and crop experts real-time information and greater control over their work.

Drones with advanced sensors, like multispectral and RGB cameras, provide clear images of crop health. Tools like NDVI (Normalized Difference Vegetation Index) help spot areas of stress, so farmers can take targeted action. Drones also create detailed soil maps that show properties like moisture levels, texture, and organic matter. This information helps farmers plan irrigation and planting better, reducing waste.

Thermal sensors on drones can detect water stress in crops and uneven watering patterns. This helps farmers adjust water use, ensuring healthier plants and saving water. Not many drones are currently being used in farming (Petkovic *et al.*, 2017) [15]. Drones are also great for quickly assessing damage from floods, droughts, or pests. The high-quality images help farmers focus on recovery and plan ways to prevent future problems.

Livestock management

Livestock management is a crucial part of farming, and new technology has led to better ways of managing animals. IoT-enabled livestock management uses technology to track and monitor animals, allowing farmers to collect real-time data about their health, environment, and behavior. This helps farmers keep an eye on their animals' movements, prevent theft, and receive alerts if any animal goes outside a designated area Two key things to manage in livestock farming are (1) the animals' quality of life and (2) the condition of the land they live on. (Akhigbe *et al.*, 2021) [3].

IoT equipment also allows farmers to continuously monitor important signs like heart rate and breathing in cattle. Automated feeding systems help manage feeding more efficiently by ensuring the right amount of food is given to each animal, reducing waste and making sure they get the proper nutrition.

Pest management

If pests or insects are not properly controlled, they can spread fast and cause serious damage. This can result in big losses for farmers, cities, and even whole countries, ruining everything from gardens to large fields and crops. To stop this, farmers use an Integrated Pest Management Program (IPMP) to control pests and protect crops. Without monitoring and controlling pests, they can harm both the economy and agriculture.

By the IoT we can use to sense the what type of the pest is attacking the crop and it will collect the data about the attack by the pest. Smart traps can help farmers use fewer harmful pesticides, which is better for people and the environment (Ahmed *et al.*, 2024) [2].

Detecting bugs and rodents through sound is another effective way to protect plant quality. Wireless sensors placed in different areas of a field can pick up the sounds made by insects. Areas with more sound waves signal a higher number of bugs. This helps farmers target pesticide spraying to those specific areas, keeping crops healthy. Smart traps can help farmers rely less on pesticides that are bad for people's health and the environment (Venkatesan *et al.*, 2018) [17]. IOT devices can detect pest activity early, allowing for timely interventions and minimizing chemical use.

IoT advantages in agriculture

There many benefits by using IOT in agriculture. many farmers were interested to implement the IOT in their farm, it reduces the yield loss and Labour cost. If the sensor detects that the soil is too dry, the farmer will be notified right away. This quick response can prevent the crop from being damaged. With the help of IoT in farming, fields can be monitored more effectively, allowing farmers to make better, data-driven decisions. IoT in farming involves using drones for precision farming. Drones can fly over big fields and take pictures to help farmers spot issues like pest problems or nutrient deficiencies. This allows farmers

to respond quickly and focus on the areas that need help, rather than treating the entire field.

Data analytics now helps farmers predict crop yields more accurately. This allows them to make better decisions about when to plant and harvest. As a result, they can increase profits, reduce waste, and help maintain a steady food supply. With IoT, farmers can track their products from the farm all the way to the consumer. In smart farming, sensors are used to collect data from different areas, such as temperature, humidity, and soil moisture. Devices like the Think to Speak IoT app, water motors, and sprinklers are also used (Gocheva *et al.*, 2021) [7]. This is especially useful today, as many consumers want to know where their food comes from. This technology allows farmers to be more transparent and share their practices with consumers, helping to build trust.

IoT helps farmers make better use of resources like water, fertilizers, and pesticides. This saves money and reduces waste, which is good for both profit and the environment. For example, automated irrigation systems can cut water use by 30-50% by adjusting based on real-time data. Fertilizers and pesticides can also be applied more efficiently, only where needed, which prevents overuse and lowers pollution.

Why farmer should practice IOT in their farm?

IOT make farmers job into ease mode and prevent them from the loses. IoT will be very responsible to the time consumption and it prevents from the time waste for the farmers. Government or middle men can be push into lose but IOT will not. IoT is a technology that helps solve the problem (Rahaman *et al.*, 2020) [14].

IoT farming helps farmers make more money, get better crop yields, and improve the environment. Using an IoT-based smart agriculture system, farmers can grow more crops of higher quality. This lowers their costs while still allowing them to earn profits. IoT in farming is growing with specialized tools, advanced software, and IT services.

In farming, the Internet of Things (IoT) is used to improve traditional methods. It helps meet increasing demand and reduce losses in production. IoT in agriculture includes robots, drones, sensors, and computer imaging. It also uses advanced tools like machine learning and analytics to monitor crops and gather data for field surveys and maps. This gives farmers valuable information to make better decisions about managing their farms, saving both time and money.

Challenges and Barriers to Adoption

Even if farmers start using IoT (Internet of Things) technology, they may not get the full benefits because many farms are in remote areas with weak or no internet access. Since farmers need to check their crop data anytime and from anywhere, a poor connection would make the whole system unreliable and almost useless. To meet the world's current needs, new ideas and technologies are always being developed and put into use (Quy,V.K *et al.*, 2022) [13]. The equipment needed for IoT in farming is expensive. While sensors are relatively cheap, putting them across all parts of a farm can still cost over a thousand dollars. Automated machines are even pricier because they include software and cloud services to store data. These tools can help farmers make more money in the long run, but the upfront cost is often too high for many to afford. IoT devices often connect to older farming equipment and use the internet, which makes them vulnerable to cyberattacks. These systems collect a lot of valuable data, and if someone hacks into the database of the IoT provider, they could steal or change the data. This makes it hard to keep the information safe and secure.

Future Prospects

In smart farming, sensors are placed throughout the fields to keep track of important environmental conditions. The Internet of Things (IoT) means connecting different devices in many ways. These devices can use sensors or RFID tags to collect information, and communication technology to send data or control things automatically. It also uses cloud computing, big data, and other modern technologies to allow remote control and analyze large amounts of data (Yue, S., *et al.*, 2021) [19]. These sensors help farmers by constantly monitoring things like soil moisture, temperature, and weather changes, ensuring crops get the right care. The smart sensors measure how much water is in the soil, helping farmers water their crops properly. This prevents overwatering or underwatering, both of which can harm plant growth. The weather stations devices collect real-time weather data, such as temperature, humidity, wind speed, and rainfall. Farmers can use this information to prepare for extreme weather events like heatwaves or storms and take preventive measures, like adjusting planting times or shielding crops. Satellites capture images of farmland from above, giving farmers a wide view of crop conditions. By analyzing these images, they can spot problems like unhealthy plants or pest issues early, allowing them to treat specific areas instead of using excessive fertilizers or pesticides. Using AI in IoT helps things run more smoothly, manage risks better, improve products and services, and make it easier to expand IoT systems (Adli, H.K *et al.*, 2023) [1]. IoT (Internet of Things) is changing the way farming works and is having a big impact on the agriculture industry. More and more farmers are using technology to improve productivity, save costs, and make food production more sustainable. The use of IoT in farming is expanding rapidly. To grow more food with limited resources, people have used major technological improvements throughout history (Khan, N *et al.*, 2021) [9]. In 2021, the market was worth $27.1 billion, and by 2031, it's expected to reach $84.5 billion. This growth is driven by the need for better farming techniques and more transparency in food production. IoT helps farmers cut costs by using resources wisely and reducing waste. In recent years, the fast growth and widespread use of the Internet of Things (IoT) has led to better practices in Wireless Sensor Networks (WSNs) for precision farming (Placidi, P *et al.*, 2021) [12]. Smart sensors analyze conditions in the field so farmers only use water, fertilizers, and pesticides when necessary, saving money and helping the environment. IoT systems will affect many areas and change how society functions as it moves into the future (Chataut, R *et al.*, 2023) [4]. Farms often struggle with labor shortages and high costs. Automated machines like self-driving tractors and robotic harvesters reduce the need for manual labor, making farming more efficient and less expensive. It's expected that by 2025, all devices will be connected to the Internet, leading to a big increase in the number of connected devices (Zikria, Y.B *et al.*, 2021) [20]. Ultimately, the biggest benefit of IoT in agriculture is that it helps farmers grow more food and improve crop quality. With better data, farmers can make smarter decisions, protect crops from risks, and increase profits.

Conclusion

In conclusion, IoT in farming is changing the way agriculture works, helping farmers use data and technology to improve their operations, boost productivity, and make farming more sustainable. By using smart devices and advanced technologies,

IoT applications allow everyone involved in farming to make better decisions, use resources more efficiently, and reduce risks in a world that's becoming more complicated. As farming continues to adopt IoT, there's a lot of potential for new ideas, growth, and positive changes in global food systems. By taking advantage of IoT's capabilities, farmers can overcome the challenges of the modern world and create a stronger, more sustainable future for both farming and society.

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